

BIRLA INSTITUTE OF TECHNOLOGY



MESRA, RANCHI
MASTERS OF COMPUTER APPLICATIONS
SESSION: MO/2025

COURSE CODE : CA410

COURSE TITLE : JAVA

JAVA LAB ASSIGNMENTS

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INDEX

SL No	Experiment	Page No	Date of Experiment	Date of Submission
1	Questions	3		
2	Assignment 1	4-6	06/11/25	07/11/25
3	Assignment 2	7-9	06/11/25	07/11/25

All Assignment Source Files are available at



<https://github.com/itzspgautam/java-lab/tree/main/Assignments>

LAB ASSIGNMENT

1. Assignment Title: Develop a simple Bank Account Management System using Java

Objective

Design a Java class BankAccount to represent a simple bank account and perform basic operations — deposit, withdraw, and check balance.

Assignment Specification

1. Define a class BankAccount with:

- Data members:
- accountNumber, accountHolderName, balance
- Constructors:
- Initialize account details and opening balance.
- Methods:
- deposit(double amount) – Adds amount to balance.
- withdraw(double amount) – Deducts amount if sufficient funds exist; otherwise display “Insufficient Balance” by throwing an exception.
- displayBalance() – Shows current balance.

2. Write a main program (BankDemo.java) to:

- Create an account object.
- Perform deposit, withdrawal, and display operations through method calls.

2. Assignment Title: Develop a Java program for a Quiz Game (MCQs).

Objective

Design a console-based multiple-choice quiz system in Java that:

- Reads questions and options from a file.
- Randomizes the question order.
- Tracks the player’s score and accuracy.

2.1 Assignment Specification

1. Input File Format (questions.txt):
2. Each question should have the following structure:

```
class BankAccount {
    private String accountNumber;
    private String accountHolderName;
    private double balance;

    // Constructor
    public BankAccount(String accountNumber, String accountHolderName, double balance) {
        this.accountNumber = accountNumber;
        this.accountHolderName = accountHolderName;
        this.balance = balance;
    }
    // Deposit method
    public void deposit(double amount) {
        balance += amount;
        System.out.println("Deposited: ₹" + amount);
    }
    // Withdraw method
    public void withdraw(double amount) throws Exception {
        if (amount > balance) {
            throw new Exception("Insufficient Balance");
        }
        balance -= amount;
        System.out.println("Withdrawn: ₹" + amount);
    }
    // Display balance
    public void displayBalance() {
        System.out.print("\n---ACCOUNT INFORMATION---");
        System.out.println("\nAccount Holder: " + accountHolderName);
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Current Balance: ₹" + balance);
    }
}

// Main Class
public class Q1_BankManagement {
    public static void main(String[] args) {

        java.util.Scanner sc = new java.util.Scanner(System.in);

        System.out.print("\n---ACCOUNT CREATION---\n");
        System.out.print("Enter Account Number: ");
        String accNo = sc.nextLine();

        System.out.print("Enter Account Holder Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Opening Balance: ");
        double bal = sc.nextDouble();

        BankAccount acc = new BankAccount(accNo, name, bal);
        System.out.println("\nAccount Created Successfully!\n");
    }
}
```

```
int choice;
do {
    System.out.println("—— Banking Menu ——");
    System.out.println("1. Deposit");
    System.out.println("2. Withdraw");
    System.out.println("3. Display Balance");
    System.out.println("4. Exit");
    System.out.print("Enter your choice: ");
    choice = sc.nextInt();

    switch (choice) {
        case 1:
            System.out.print("Enter amount to deposit: ");
            double dep = sc.nextDouble();
            acc.deposit(dep);
            break;

        case 2:
            System.out.print("Enter amount to withdraw: ");
            double wd = sc.nextDouble();
            try {
                acc.withdraw(wd);
            } catch (Exception e) {
                System.out.println(e.getMessage());
            }
            break;

        case 3:
            acc.displayBalance();
            break;

        case 4:
            System.out.println("Exiting ... ");
            break;

        default:
            System.out.println("Invalid Choice!");
    }
    System.out.println();
} while (choice != 4);
sc.close();
}
```

● itzspgautam@Surajs-MacBook-Pro Assignments % java Q1_BankManagement

Q1: OUTPUT

---ACCOUNT CREATION---

Enter Account Number: 10011

Enter Account Holder Name: Suraj Prakash Gautam

Enter Opening Balance: 1000

Account Created Successfully!

----- Banking Menu -----

1. Deposit

2. Withdraw

3. Display Balance

4. Exit

Enter your choice: 1

Enter amount to deposit: 500

Deposited: ₹500.0

----- Banking Menu -----

1. Deposit

2. Withdraw

3. Display Balance

4. Exit

Enter your choice: 3

---ACCOUNT INFORMATION---

Account Holder: Suraj Prakash Gautam

Account Number: 10011

Current Balance: ₹1500.0

----- Banking Menu -----

1. Deposit

2. Withdraw

3. Display Balance

4. Exit

Enter your choice: 2

Enter amount to withdraw: 200

Withdrawn: ₹200.0

----- Banking Menu -----

1. Deposit

2. Withdraw

3. Display Balance

4. Exit

Enter your choice: 3

---ACCOUNT INFORMATION---

Account Holder: Suraj Prakash Gautam

Account Number: 10011

Current Balance: ₹1300.0

----- Banking Menu -----

1. Deposit

2. Withdraw

3. Display Balance

4. Exit

Enter your choice: 4

Exiting...

Question.java

Q2: CODE

```

public class Question {
    String questionText;
    String[] options;
    String correctAnswer;

    public Question(String questionText, String[] options, String correctAnswer) {
        this.questionText = questionText;
        this.options = options;
        this.correctAnswer = correctAnswer;
    }

    public boolean isCorrect(String userAnswer) {
        return userAnswer.equalsIgnoreCase(correctAnswer);
    }
}

```

questions.txt

Q2: CODE

```

Which language is platform independent?
C
C++
Java
Python
C
Who developed Java?
Dennis Ritchie
James Gosling
Guido van Rossum
Bjarne Stroustrup
B
Which company currently owns Java?
Sun Microsystems
Oracle
Microsoft
Google
B
Which of the following is NOT a Java feature?
Object Oriented
Platform Independent
Pointer Arithmetic
Automatic Memory Management

```

```

import java.io.BufferedReader;
import java.io.FileReader;
import java.util.ArrayList;
import java.util.Collections;
import java.util.List;
import java.util.Scanner;

public class QuizGame {
    public static void main(String[] args) {
        List<Question> questions = new ArrayList<>();
        // Reading questions from file
        try {
            BufferedReader br = new BufferedReader(new FileReader("questions.txt"));
            String line;
            while ((line = br.readLine()) != null) {
                String q = line;
                String a = br.readLine();
                String b = br.readLine();
                String c = br.readLine();
                String d = br.readLine();
                String correct = br.readLine();

                String[] options = { a, b, c, d };
                questions.add(new Question(q, options, correct));
            }
            br.close();
        } catch (Exception e) {
            System.out.println("Error reading file: " + e.getMessage());
            return;
        }
        // Shuffle questions
        Collections.shuffle(questions);

        System.out.println("Welcome to the Java Quiz!\n");

        Scanner sc = new Scanner(System.in);
        int total = questions.size();
        int correctCount = 0;
        int qNo = 1;

        for (Question q : questions) {
            System.out.println("Q" + qNo + ". " + q.questionText);
            System.out.println("A. " + q.options[0]);
            System.out.println("B. " + q.options[1]);
            System.out.println("C. " + q.options[2]);
            System.out.println("D. " + q.options[3]);
            System.out.print("Enter your answer (A/B/C/D): ");
            String ans = sc.nextLine();

            if (q.isCorrect(ans)) {
                System.out.println("Correct!\n");
                correctCount++;
            } else {
                System.out.println("Wrong! Correct answer: " + q.correctAnswer + "\n");
            }
            qNo++;
        }
        double accuracy = (correctCount * 100.0) / total;
        System.out.println("You scored " + correctCount + " out of " + total);
        System.out.println("Accuracy: " + accuracy + "%");

        sc.close();
    }
}

```



```
● itzspgautam@Surajs-MacBook-Pro A2 % javac Question.java QuizGame.java
● itzspgautam@Surajs-MacBook-Pro A2 % java QuizGame
Welcome to the Java Quiz!
```

Q2: OUTPUT

Q1. Which company currently owns Java?

- A. Sun Microsystems
- B. Oracle
- C. Microsoft
- D. Google

Enter your answer (A/B/C/D): B

Correct!

Q2. Which of the following is NOT a Java feature?

- A. Object Oriented
- B. Platform Independent
- C. Pointer Arithmetic
- D. Automatic Memory Management

Enter your answer (A/B/C/D): B

Wrong! Correct answer: C

Q3. Who developed Java?

- A. Dennis Ritchie
- B. James Gosling
- C. Guido van Rossum
- D. Bjarne Stroustrup

Enter your answer (A/B/C/D): B

Correct!

Q4. Which language is platform independent?

- A. C
- B. C++
- C. Java
- D. Python

Enter your answer (A/B/C/D): C

Correct!

You scored 3 out of 4

Accuracy: 75.0%

```
● itzspgautam@Surajs-MacBook-Pro A2 % █
```



Thank
You

SURAJ PRAKASH GAUTAM

